Maven Kim

mavenkim2@gmail.com | mavenkim2.github.io | linkedin.com/in/maven-kim | github.com/mavenkim2

Education

University of Pennsylvania Expected: May 2028
Master of Science in Engineering in Computer Graphics and Game Technology
University of Virginia Graduated: May 2023
Bachelor of Science in Computer Science GPA: 3.916

Technical Skills

Languages: C++, C, HLSL, Python, C#
Technologies: CUDA, OptiX, Vulkan, OpenUSD, MaterialX, Nsight, Intel VTune, CMake, Git

Projects

Yeoubi - GPU Path Tracer | C++, CUDA, OptiX Jun 2026 - Present

- Built a physically based wavefront path tracer using CUDA and NVIDIA OptiX
- Implemented GPU path guiding on surfaces with parallax-aware von Mises-Fisher mixture fitting and illumination-aware spatial subdivision
- Optimized CUDA path-guiding kernels to process 85 million training samples in approximately 5 seconds per update
- Developed an OpenUSD scene pipeline supporting instancing and MaterialX shading networks

Interactive Rendering of the Moana Island Scene | C++, Vulkan Mar 2025 - Oct 2025

- Rendered the Moana island scene at 20-30 FPS on an RTX 4070 Laptop GPU with 8 GB VRAM
- Developed a faster decoding algorithm for AMD's Dense Geometry Format, reducing frame time by 14%
- Reduced GPU memory footprint by 1.5 GB using RTX Mega Geometry and triangle mesh compression
- Implemented real-time streaming of 23 GB of Ptex texture assets

Monte Carlo Path Tracer Jul 2024 - Mar 2025

- Implemented a quasi-Monte Carlo physically based path tracer from scratch in C++
- Rendered adaptively tessellated Catmull-Clark subdivision surfaces with displacement mapping using the open source libraries OpenSubdiv and Ptex
- Optimized execution with SIMD intersection tests and hit point sorting for better texture cache coherency
- Added support for scenes with millions of instances using multi-level instancing and partial rebraiding
- Rendered the Moana island scene at 1920 x 804 with 64 samples per pixel in 213 seconds on a 24-core Intel processor

Work Experience

GEICO Technology Solutions Chevy Chase, MD

Technology Development Program – Software Engineer Development Track (SDP) Jun 2023 - Jun 2024

- Migrated the underwriting systems codebase from MuleSoft to Azure Functions, reducing operating costs
- Implemented a ZIP-code lookup API for a mock sales application, mapping customer ZIP codes to city and state

Amazon Arlington, VA

Software Development Engineer Intern - AWS Cryptography Team May 2022 - Aug 2022

- Built an OPAQUE password-authentication prototype for Amazon Ring using Python, Docker, and AWS Nitro Enclaves
- Designed and implemented cryptographic secrets storage for user registration and authentication using AWS KMS and DynamoDB
- Automated prototype setup using a deployment script and AWS CloudFormation